



EmotionCam User Manual

Local visible-expression estimates with privacy-first personalized calibration

Version 1.0.0 | Updated June 15, 2026

1. Cover Page



EmotionCam

Local visible-expression estimates, with privacy-first personalized calibration

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EmotionCam estimates visible facial expressions. It does not know or guarantee a person's true emotions.

2. Introduction

EmotionCam is a Windows desktop app for one person using a laptop webcam. It processes the live preview locally, estimates visible facial expressions, and presents a smoothed label, group, confidence estimate, interaction message, and timeline.

The app can learn the user's visible facial movements through personalized calibration, run locally in the Windows system tray, and optionally save metadata-only expression history.

3. Privacy and Safety

- Webcam processing is local. The current implementation has no cloud calls.
- EmotionCam does not identify people or perform face matching.
- There is no account, telemetry, analytics, automatic screenshot, or video-recording feature.
- Optional logs contain timestamps, visible-expression labels/groups, confidence, status, messages, popup status, and FPS.
- Calibration stores normalized landmark feature vectors and labels locally.
- Raw calibration frames are not saved unless Store raw calibration images for debugging is explicitly enabled.
- The disabled External AI agent option is a future placeholder and cannot send frames.

Do not use EmotionCam for medical, psychological, legal, hiring, security, surveillance, or lie-detection decisions.

4. Installation

Install the packaged app

- 1 Open EmotionCam-Setup-1.0.0.exe.
- 1 Complete the per-user installer.
- 1 Choose the optional desktop shortcut if desired.
- 1 Launch EmotionCam from the Start Menu or desktop shortcut.

The installer uses the simplified smiling camera-lens icon for the installer, executable, shortcuts, taskbar, and tray.

Run from source

```
cd "C:\path\to\AI webcam emotion detector\emotioncam"
```

```
py -3.12 -m venv .venv
```

```
.\.venv\Scripts\Activate.ps1
```

```
python -m pip install --upgrade pip
```

```
python -m pip install -r requirements.txt
```

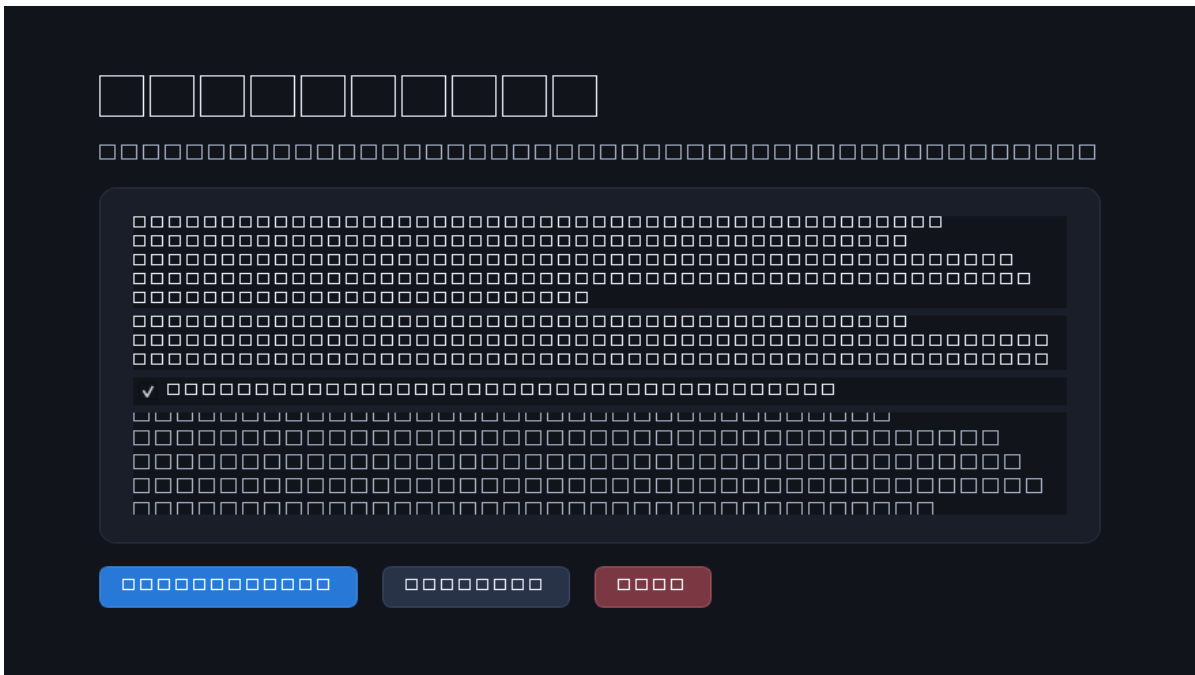
```
python -m app.main
```

Allow camera access for desktop apps in Windows Settings. Close Camera, Teams, Zoom, browser calls, and other apps using the webcam.

5. First Launch

The startup screen shows the local-processing privacy notice and approximation notice. Save local expression metadata history is enabled by default but can be disabled before starting.

- Start Camera saves the logging choice, opens the dashboard, and starts the webcam.
- Settings opens configuration before starting.
- Exit closes EmotionCam.
- Help > Start Demo Guide opens the offline demo page.



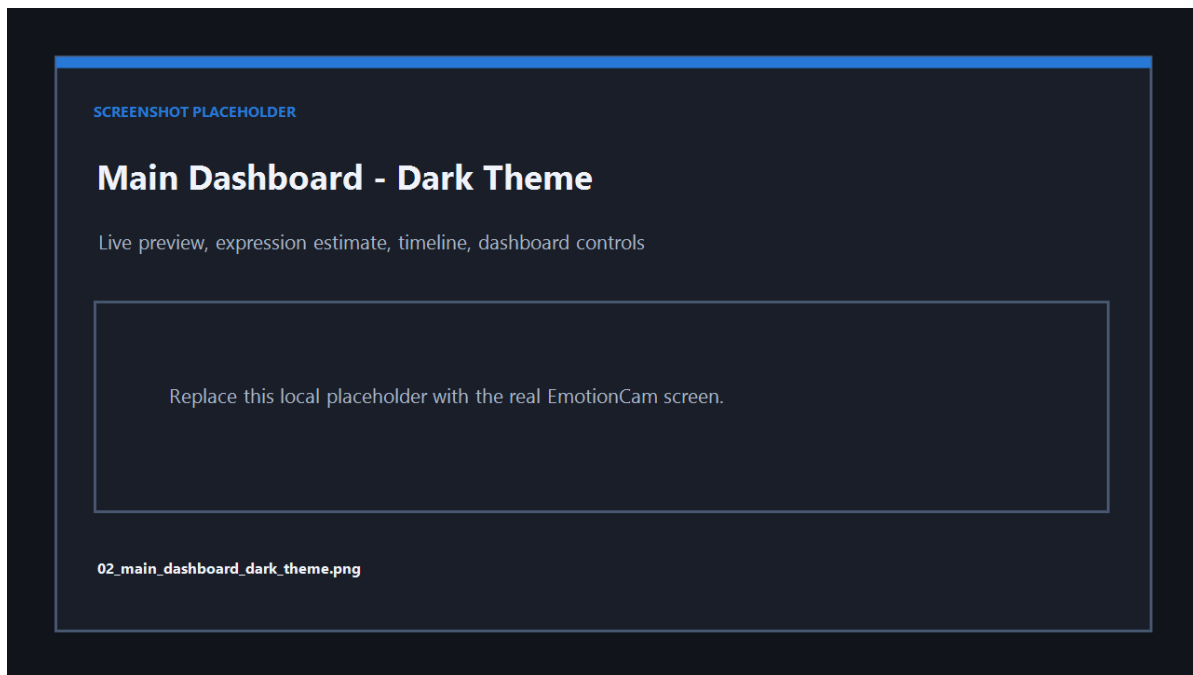
Placeholder screen example: capture the real startup screen as docs/screenshots/01_startup_screen.png.

6. Main Dashboard

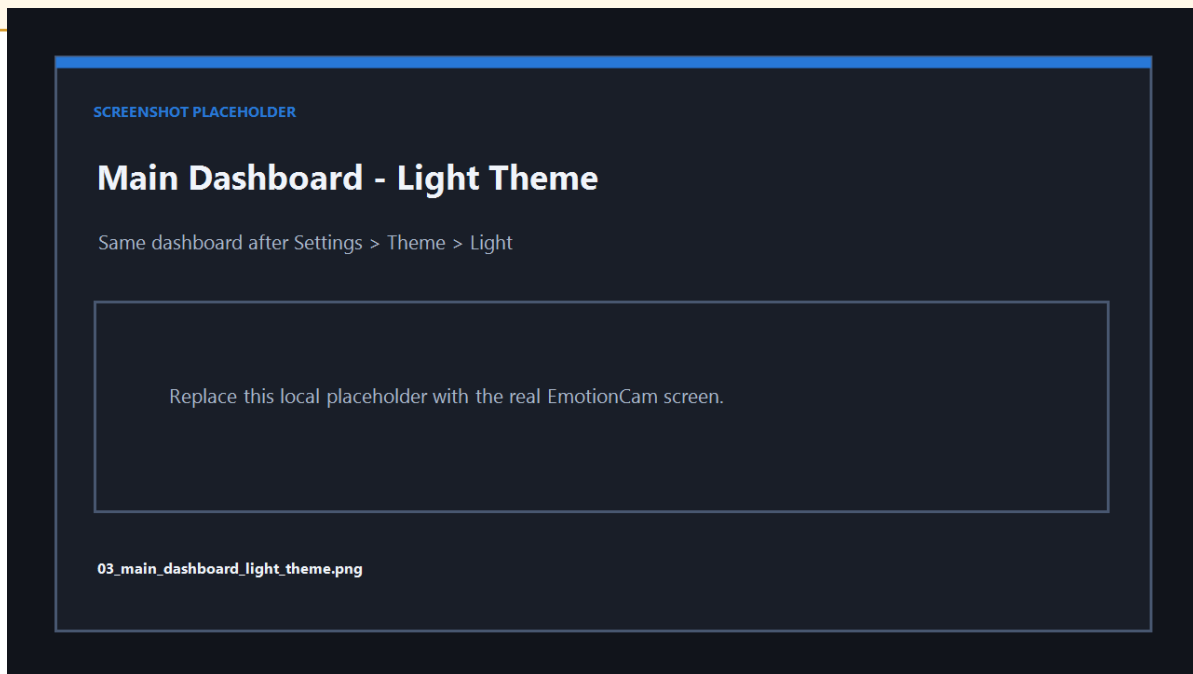
Area or control	Purpose
Webcam preview	Live local preview and optional face rectangle
Current likely expression	Smoothed visible-expression estimate
Expression group / confidence	Positive, negative, or neutral group and estimated confidence
Interaction message / timeline	Safe text message and recent smoothed labels
Detection mode / profile status	Heuristic, Personalized, or Hybrid; profile active/inactive
Start Camera / Stop Camera	Opens or releases the webcam
Pause Analysis	Keeps preview active while pausing classification
Settings / Open Logs	Opens configuration or the local logs folder
Run in background	Keeps local analysis active in tray/taskbar mode
Toggle Debug	Shows raw/smoothed estimates and detector details
Improve detection with calibration	Opens personalized calibration
Exit	Stops the camera and closes the app

Rectangle colors use the smoothed group:

- Blue: positive
- Orange: negative
- Gray: neutral, unknown, low confidence, or no face



Placeholder: dark dashboard.

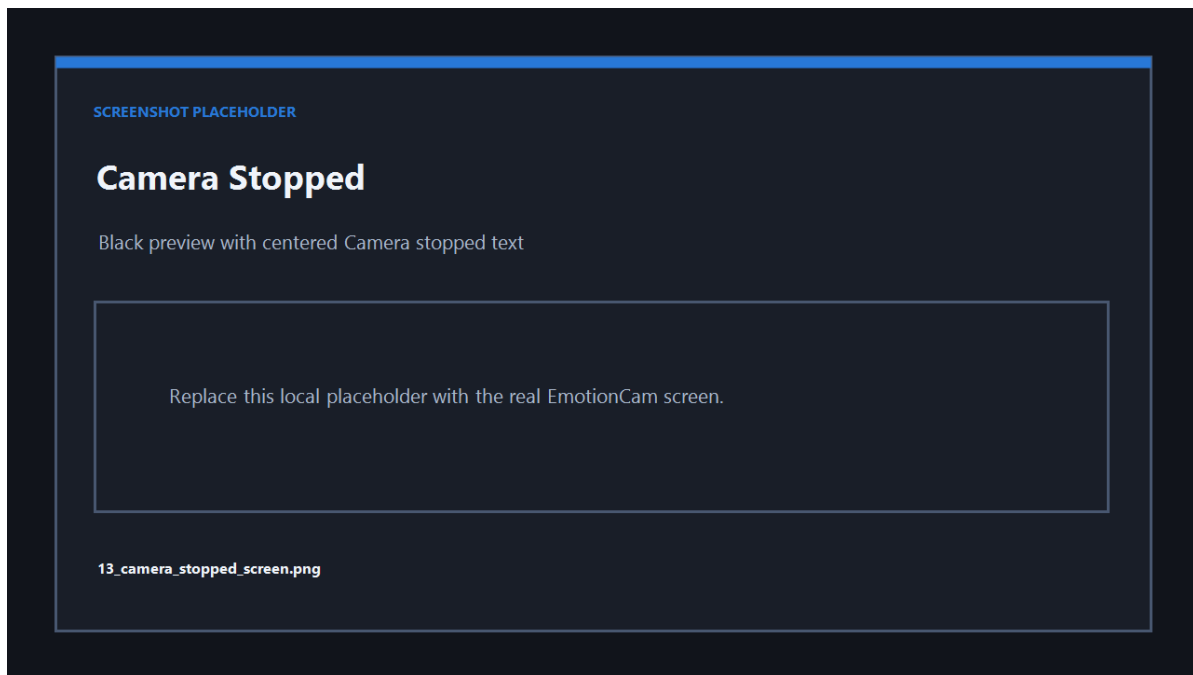


Placeholder: light dashboard after changing Settings > Theme > Light and saving.

7. Starting and Stopping Camera

- 1 Select Start Camera.
- 1 Wait for Camera running - analysis is local.
- 1 Use Pause Analysis to pause classification while the preview remains active.
- 1 Use Stop Camera to stop analysis, release the webcam, and remove the last frame.

- 1 Confirm the preview is black with centered Camera stopped.
- 1 Select Start Camera to restart.



Placeholder: black stopped-camera screen.

8. Expression Detection

Heuristic labels include neutral, happy, smile, laughing, surprised, sad, angry, fearful, disgusted, confused, focused, tired, unknown, low_confidence, and no_face.

Personalized calibration additionally supports smile_small, smile_big, amused, bored, frustrated, concerned, skeptical, thinking, and relaxed.

Group	Labels
Positive	happy, smile, smile_small, smile_big, laughing, amused, surprised
Negative	sad, angry, fearful, disgusted, tired, bored, frustrated, concerned
Neutral	neutral, focused, confused, skeptical, thinking, relaxed, unknown, low_confidence, no_face

Confidence is an estimate, not a guarantee. Temporal smoothing and a stable-expression duration reduce flicker. Strong negative labels require more confidence, while uncertain results fall back toward neutral or unknown.

Lighting, blur, distance, glasses, occlusion, and head angle can affect results. MediaPipe landmarks improve moderate-angle handling; OpenCV is a fallback. Calibration helps because it learns local normalized feature patterns from the user's own visible movements.

9. Personalized Calibration

Open calibration from Improve detection with calibration or Settings > Train / Calibrate Expressions.

The unified calibration screen includes:

- Target-expression dropdown

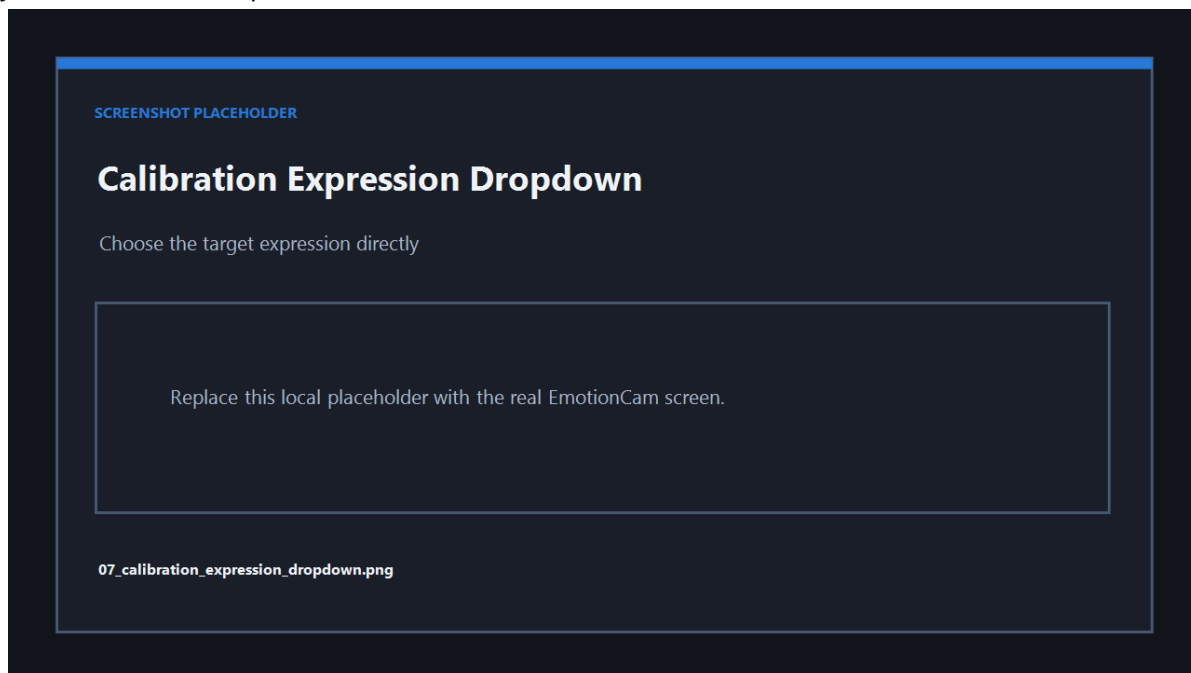
- Local example expression graphic
- Live webcam preview
- Live detected-expression estimate
- Match helper/checkmark
- Previous, Start Capture, Re-capture, Next, and Finish
- Capture progress, valid/rejected/stored counts, rejection reason, and quality summary

The match helper shows when the current smoothed estimate roughly matches the target or a related expression family. It is guidance only and never blocks capture. Example graphics are local guides and do not need to be copied perfectly.

Capture an expression

- 1 Choose a target from the dropdown.
- 1 Prepare the visible expression.
- 1 Select Start Capture.
- 1 Review valid/rejected counts and quality.
- 1 Select Re-capture to remove and replace that target's previous batch.
- 1 Navigate with Previous, Next, or the dropdown.
- 1 Select Finish after at least one valid target is stored.

The old Skip button and separate Label Unknown Expression flow were removed. Users now choose the target directly from the unified dropdown.



SCREENSHOT PLACEHOLDER

Calibration Example Graphic

Bundled offline visual guide for the target

Replace this local placeholder with the real EmotionCam screen.

08_calibration_example_graphic.png

SCREENSHOT PLACEHOLDER

Calibration Match Helper

Permissive match indicator; capture remains allowed

Replace this local placeholder with the real EmotionCam screen.

09_calibration_match_checkmark.png

SCREENSHOT PLACEHOLDER

Calibration Capture Progress

Progress, valid/rejected samples, and quality

Replace this local placeholder with the real EmotionCam screen.

10_calibration_capture_progress.png

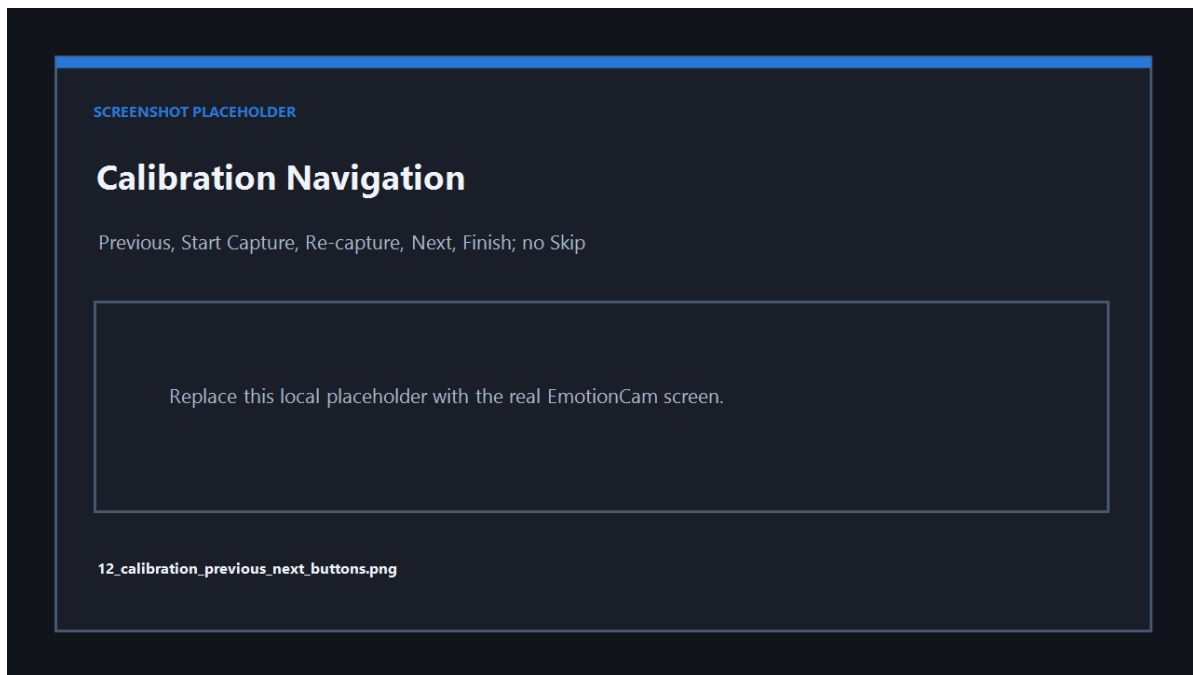
SCREENSHOT PLACEHOLDER

Calibration Re-capture

Re-capture replaces the selected target batch

Replace this local placeholder with the real EmotionCam screen.

11_calibration_recapture_button.png



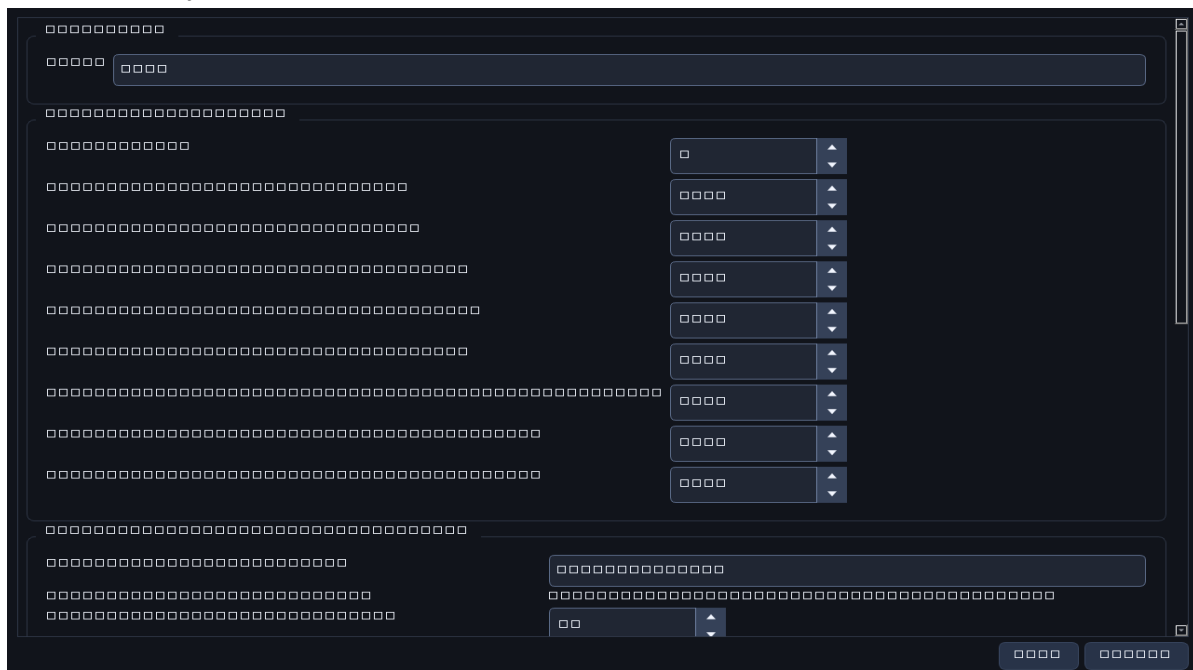
These are placeholder references until real screenshots are captured.

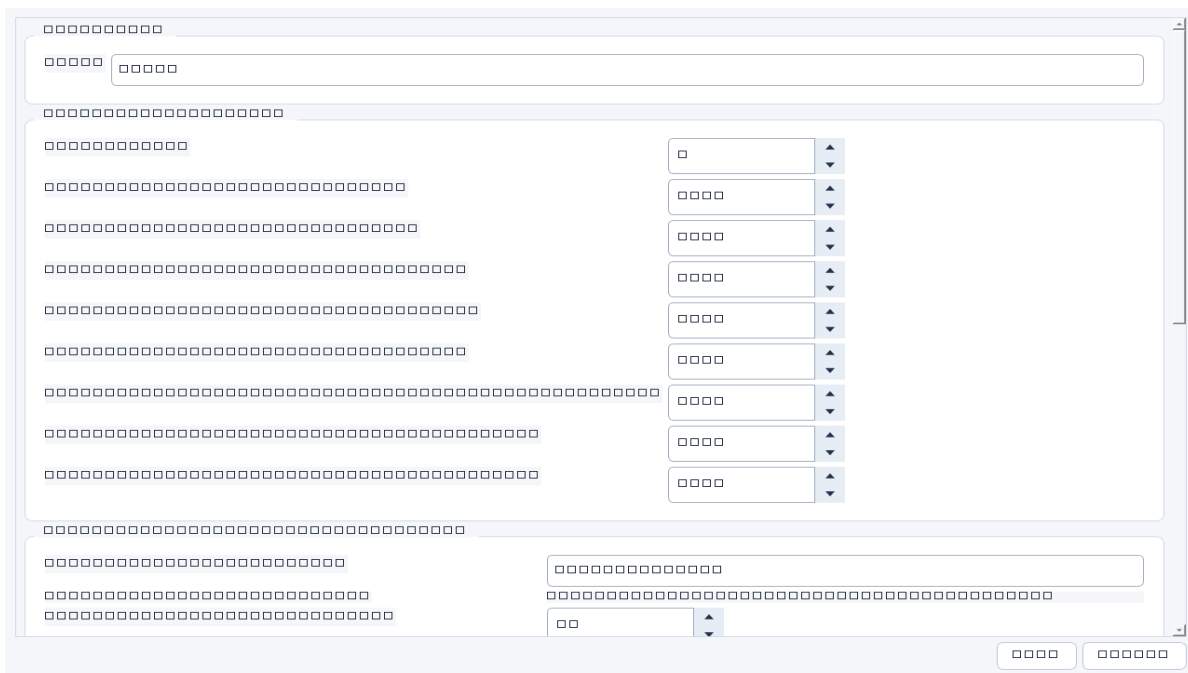
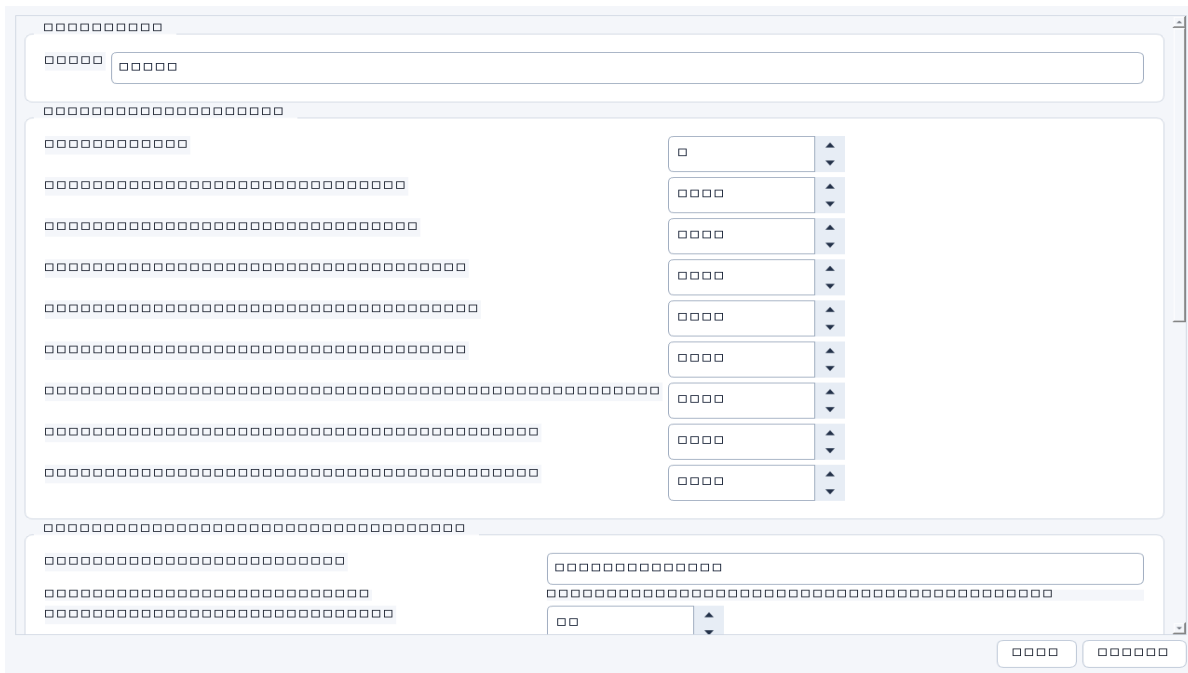
10. Settings

Appearance and numeric controls

Use Settings > Appearance > Theme to select Dark or Light, then select Save. The theme choice is saved locally and applied app-wide to the startup screen, dashboard, settings, calibration, menus, buttons, panels, inputs, dropdowns, progress bars, and messages.

All numeric fields are spinner controls. Type a value or use the visible up/down arrow buttons. The arrow area is wide, clickable, and styled for both themes.





Placeholder screen examples: capture Settings in both themes with the spinner arrow area visible.

Actual defaults

Setting / config key	Default
Theme (theme)	dark
Camera index (camera_index)	0
Detection confidence (detection_confidence_threshold)	0.45
Expression confidence (expression_confidence_threshold)	0.45

Setting / config key	Default
Smoothing window (smoothing_window_seconds)	1.5 s
Stable expression (stable_expression_seconds)	0.8 s
Message cooldown (message_cooldown_seconds)	8 s
Face-missing grace (face_missing_grace_seconds)	1.0 s
Negative minimum confidence	0.65
Positive minimum confidence	0.50
Show face rectangle / confidence / messages	On
Local metadata logging	On
Background popups	On
Negative popup cooldown	600 s
Positive recovery cooldown	300 s
Debug panel	Off
Detection mode	heuristic initially; Hybrid after calibration
Personalized profile enabled	On
Calibration target samples	45
Calibration capture duration	4.0 s
Calibration preparation countdown	2.0 s stored; current capture is manual
Minimum good samples	15
Store raw calibration images	Off
External AI backend	Off and disabled

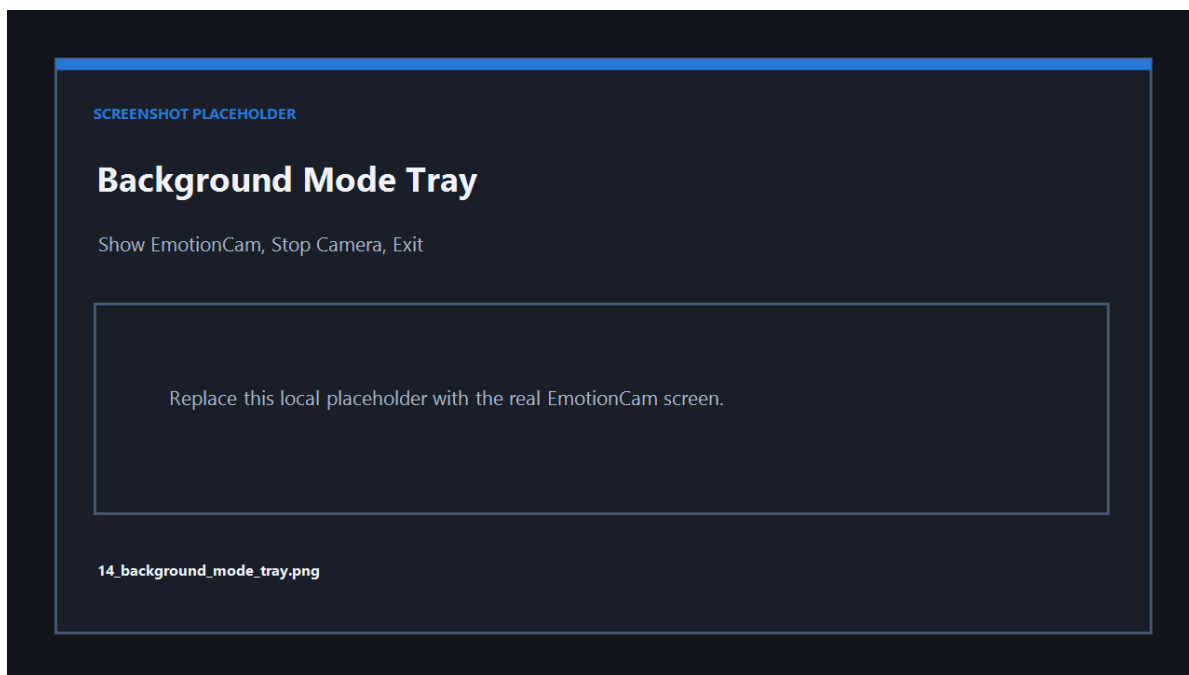
Settings also provide local profile train/retrain/add/delete/export/import controls, log controls, and reset-to-default.

11. Background Mode

Select Run in background to keep camera analysis and optional metadata logging active locally. EmotionCam hides to the Windows tray when available, or minimizes to the taskbar.

The tray menu contains Show EmotionCam, Stop Camera, and Exit. Double-click the tray icon to restore the app.

With background popups enabled, a stable negative group can show a cooldown-controlled supportive notification. A later stable positive group can show a positive-recovery notification. These are expression estimates, not diagnoses.



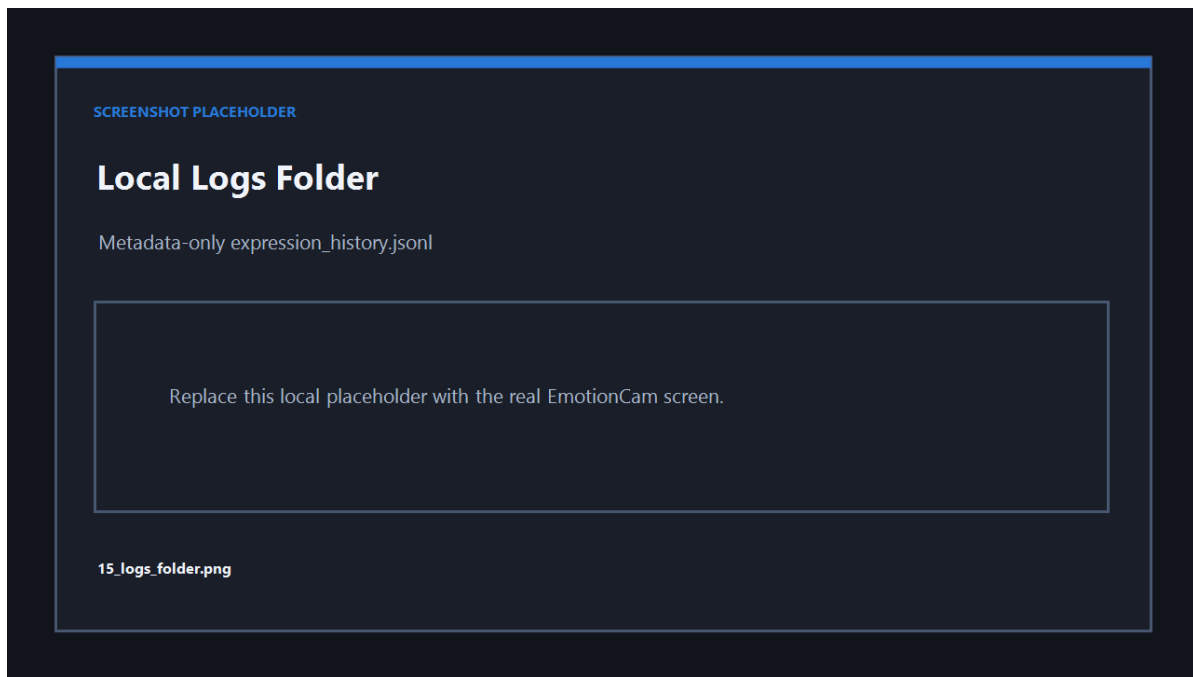
Placeholder: tray icon and menu.

12. Logs and Local Data

Data	Actual path
Configuration	%LOCALAPPDATA%\EmotionCam\config.json
Metadata history	%LOCALAPPDATA%\EmotionCam\logs\expression_history.jsonl
Personalized profile	%LOCALAPPDATA%\EmotionCam\profile\expression_profile.json
Optional debug calibration frames	%LOCALAPPDATA%\EmotionCam\profile\debug_images\

The JSONL log stores timestamp, expression label/group, confidence, detection mode, classifier source, profile-active status, face-detected status, displayed message, popup status, and FPS. It does not save frame/image fields or identity data.

Use Open Logs, Clear Logs, or disable local logging in Settings. Use Settings to retrain, export, import, or delete the personalized profile.



Placeholder: local logs folder.

13. Complete Demo Walkthrough

Start from Help > Start Demo Guide, or open docs\START_DEMO_HERE.html.

#	What to click/do	What should appear	Expected result / screenshot
1	Launch EmotionCam	Simplified lens icon and startup screen	01_startup_screen.png
2	Point out the icon	Same icon in app/taskbar	Branding is consistent
3	Review startup privacy text	Logging choice and local-processing notice	No camera yet
4	Select Start Camera	Dark dashboard and preview	02_main_dashboard_dark_theme.png
5	Hold neutral, then smile	Estimate changes; positive group turns rectangle blue	02_main_dashboard_dark_theme.png
6	Open Settings	Scrollable settings dialog	Settings visible
7	Click numeric up/down arrows and type a value	Spinner changes both ways	04_settings_numeric_arrows_dark.png
8	Choose Theme: Light and Save	Entire app becomes light	03_main_dashboard_light_theme.png
9	Optionally switch back to Dark	Entire app returns dark	06_settings_theme_toggle.png
10	Open calibration	Unified calibration screen	07_calibration_expression_dropdown.png
11	Select an expression from dropdown	Target changes	07_calibration_expression_dropdown.png
12	Point out local example graphic	Graphic changes with target	08_calibration_example_graphic.png
13	Perform a roughly matching expression	Helper match may appear	09_calibration_match_checkmark.png
14	Select Start Capture	Local feature capture starts	10_calibration_capture_progress.png
15	Show capture progress	Counts and quality update	10_calibration_capture_progress.png
16	Select Re-capture	Current target batch is removed/reset	11_calibration_recapture_button.png

#	What to click/do	What should appear	Expected result / screenshot
17	Use Previous and Next	Target moves in both directions	12_calibration_previous_next_buttons.png
18	Capture valid samples and Finish	Completion message/profile saved	Profile activates
19	Return to dashboard	Hybrid mode/profile active	Dashboard status
20	Select Run in background	Tray icon/menu; analysis continues locally	14_background_mode_tray.png
21	Restore app and select Open Logs	Metadata-only logs folder	15_logs_folder.png
22	Select Exit	Camera released and app closes	Demo complete

See START_DEMO_HERE.html and demo_script.md.

14. Troubleshooting

Issue	What to try
Camera will not start	Allow desktop-app camera access and close other webcam apps.
No face detected	Improve lighting, face the camera, move closer, and remove occlusion.
Estimate is wrong	Hold steady, reduce head angle, enable Debug, and calibrate.
Too few calibration samples	Improve light/position, hold still, or increase capture duration.
Spinner arrows do not appear	Restart the updated app/build; confirm the current theme applies and no old executable is running.
Theme does not change	Select the theme and then **Save** ; the choice is stored in local config.
Background popup does not show	Enable popups, allow Windows notifications, and wait for cooldown.
Logs are missing	Enable local metadata logging and run analysis briefly.
Installer is blocked	Verify the source before using SmartScreen **More info > Run anyway** .
Source dependencies are missing	Activate .venv and run <code>python -m pip install -r requirements.txt</code> .

15. Limitations

- Visible-expression estimates are approximate and can be wrong.
- Lighting, blur, occlusion, distance, and head angle affect results.
- Personalized calibration improves estimates but does not guarantee accuracy.
- The app is designed for one user and one laptop webcam.
- The match helper is permissive guidance and does not validate a true emotion.
- The preparation-countdown config remains, but capture currently starts manually.
- EmotionCam does not know true feelings and is not for high-stakes decisions.

16. Uninstall and Data Removal

Uninstall from Windows Settings > Apps > Installed apps > EmotionCam > Uninstall.

To remove all remaining local settings, logs, profile features, and optional debug calibration frames, fully exit EmotionCam and delete:

%LOCALAPPDATA%\EmotionCam

17. How to Start the Demo

- 1 Install or run EmotionCam.
- 1 Open the app.
- 1 Select Help > Start Demo Guide.
- 1 Follow the offline docs\START_DEMO_HERE.html page.

Directly opening docs\START_DEMO_HERE.html also works without internet access.